

Morning Paper Report - Tuesday, July 07, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Enabling Human-like Language-Capable Robots Through Working Memory Modeling

Rafael Sousa Silva, Tom Williams

IEEE/ACM International Conference on Human-Robot Interaction, 2023

Source: Semantic Scholar; matched terms: working memory, modeling, working memory modeling

Link: <https://www.semanticscholar.org/paper/164da273b6c2f58cb38653ab80b536cdde514a95>

Goal: Working Memory (WM) is a central component of cognition

Method: Thus, for robots to achieve human-like natural language capabilities, we argue that their cognitive models should include an accurate WM representation that plays a similarly central role

Results: Our research investigates how different WM models from cognitive psychology affect robots' natural language capabilities Specifically, we explore the limited capacity nature of WM and how different information forgetting strategies, namely decay and interference, impact the human-likeness of utterances formulated by robots.

Conclusion: Our research investigates how different WM models from cognitive psychology affect robots' natural language capabilities Specifically, we explore the limited capacity nature of WM and how different information forgetting strategies, namely decay and interference, impact the human-likeness of utterances formulated by robots.

2. Short Report: Environmental Sensitivity in Children is Associated with Visual Working Memory and Visual Search Performance

Jenni Elise Kähkönen, Francesca Lionetti, Michael Pluess

2025

Source: Crossref; matched terms: visual working memory, visual search, working memory

Link: https://doi.org/10.31219/osf.io/k6fg7_v1

Goal: Highly sensitive children perceive and process stimuli more deeply but the developmental mechanisms enabling this deeper processing are not widely understood yet

Method: Ninety-seven 7–9-year-old ($M = 7.99$, $SD = 0.34$) UK primary school children (47% girls) completed tasks on visual working memory and visual search Teachers and children reported on children's environmental sensitivity

Results: Only child-rated sensitivity positively predicted visual working memory performance, with less sensitive children scoring particularly low

Conclusion: Visual search findings suggested that highly sensitive children make fewer impulsive mistakes, aligning with theory and research

3. Modeling Working Memory with Convolutional Neural Networks for Knowledge Tracing

Tao Huang, Bin Chen, Huali Yang, Jing Geng, Hekun Xie, Junjie Hu

SSRN Electronic Journal, 2022

Source: Crossref; matched terms: working memory, modeling, neural network, neural networks

Link: <https://doi.org/10.2139/ssrn.4111801>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

4. Emotional content effects the precision of visual working memory

Zeinab Haghian, Abdol-Hossein Vahabie, Ehsan Rezayat
2024

Source: Crossref; matched terms: visual working memory, working memory, precision

Link: <https://doi.org/10.1101/2024.12.01.626217>

Goal: Abstract Objectives Our working memory (WM) is susceptible to errors influenced by various sources

Method: Participants were tasked with recalling the emotional valence of target faces across various n-back conditions, selecting their responses from a continuum of 19 morphed faces ranging from sad to happy

Results: Results Our findings indicate that participants generally exhibit a negativity bias, struggling more with happy faces

Conclusion: Results Our findings indicate that participants generally exhibit a negativity bias, struggling more with happy faces

5. Aging Decreases the Precision of Visual Working Memory

Shahzad Mohammadpour Esfahan, Mohammad-Hossein H.K Nili, Mehdi Sanayei, Ehsan Rezayat
2023

Source: Crossref; matched terms: visual working memory, working memory, precision

Link: <https://doi.org/10.1101/2023.01.03.522567>

Goal: Abstract Objectives Working memory (WM) is a cognitive ability that enables us to hold information temporarily

Method: Results The error of participants increased significantly as they aged

Results: Results The error of participants increased significantly as they aged

Conclusion: The results also indicated that the probability of guessing the response as a measure of random response is not affected by age.